Week 3: Logistic Functions and Bounded Growth

Theme: "When Growth Has Limits"

Science Content: Fish populations, carrying capacity, sustainable harvesting, the Schaefer model

Learning Outcomes: At the end of this week you should be able to: 1. Understand why exponential growth cannot continue indefinitely in real ecosystems 2. Recognize the logistic function as a model for growth with limits 3. Identify carrying capacity and intrinsic growth rate from data 4. Calculate Maximum Sustainable Yield (MSY) for renewable resources 5. Understand arithmetic sequences and their connection to linear functions 6. Compare functions and find intersections

Exam Alignment: AP: AA1, QF

3.1 Introduction: Why Growth Can't Continue Forever

The Problem with Unlimited Growth

In Week 2, we studied exponential growth: $P(t) = P_0e^{rt}$. This model captures the early stages of population growth beautifully—bacteria in a petri dish, yeast in a fermentation tank, a few rabbits in a new meadow. But it cannot last forever.

Ask: Can this exponential pattern continue forever? What would happen if it did?

The answer is clearly no. Consider:

Scenario	Limiting Factor
Bacteria in petri dish	Nutrients exhausted, waste accumulates
Fish in the ocean	Food competition, predators, habitat space
Rabbits in meadow	Food supply, predators, disease, space
Human population	Health systems, resources, pollution, space

Key Insight: All real growth patterns eventually encounter **constraining factors** that slow and eventually halt growth. The exponential model is useful only when the population is small relative to its environment.

From Stock to Flow: A Reminder from Prerequisites

From 6th-grade science (Stock and Flow Unit), we learned that:

- **Stock** (S): The total amount at a point in time (e.g., 2 million tonnes of fish)
- **Flow** or **Growth** (F): The change in stock over time (e.g., 200 tonnes of additional fish per year)

The fundamental question becomes: **How does the flow (growth) depend on the level of the stock?**

3.2 The Logistic Function

Building Intuition

If fish grow at a constant rate r regardless of population size:

$$G(S) = rS$$

Then the stock would grow exponentially: $S(t) = S_0e^{rt}$

But in the real world, **competition for resources increases**. The actual growth rate must decline as the population approaches the environment's carrying capacity K.

The Logistic Growth Model

The logistic function incorporates this constraint (degrade):

$$P(t) = \frac{C}{1 + Ae^{-rt}}$$

where: C = **carrying capacity** (maximum sustainable population), r = **intrinsic growth rate** (growth rate when population is small), A = parameter related to initial conditions: $A = \frac{C - P_0}{P_0}$, P_0 = initial population

Alternative Form

The logistic function is sometimes written with an inflection point parameter:

$$P(t) = \frac{C}{1 + e^{-r(t-t_0)}}$$

where: L = carrying capacity, k = growth rate parameter, t_0 = time at inflection point (where growth rate is maximum)

Properties of the Logistic Curve (S-Curve)

Property	Value/Description
Domain	$(-\infty, \infty)$
Range	$(0, C)$ or $(0, K)$
Initial value	$P(0) = \frac{C}{1+A}$
As $t \to \infty$	$P(t) \to K$ (horizontal asymptote)
As $t \to -\infty$	$P(t) \to 0$ (horizontal asymptote)
Inflection point	$P = K/2$, where growth rate is maximum
Shape	S-curve (sigmoid)

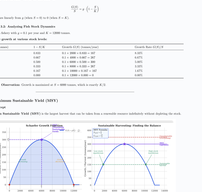

Figure 1: Logistic Growth

Example 3.1: Population with Carrying Capacity

A fish population follows the logistic equation:

$$P(t) = \frac{1500}{1 + 2.5e^{-0.3t}}$$

where $P(t)$ is in tonnes and t is in years.

Questions: 1. What is the carrying capacity? 2. What is the initial population? 3. How long until the population reaches 50% of carrying capacity?

Solution:

1. Carrying capacity: $K = 1500$ tonnes
2. Initial population:

$$P(0) = \frac{1500}{1 + 2.5e^{0}} = \frac{1500}{3.5} = 428.6 \text{ tonnes}$$

3. Time to reach $P = 750$ tonnes:

$$750 = \frac{1500}{1 + 2.5e^{-0.3t}}$$
$$1 + 2.5e^{-0.3t} = 2$$
$$2.5e^{-0.3t} = 1$$
$$e^{-0.3t} = 0.4$$
$$-0.3t = \ln(0.4)$$
$$t = \frac{\ln(0.4)}{-0.3} = 3.05 \text{ years}$$

Note: This matrices Qt in the exam, which asks about time to reach 50% of carrying capacity.

3.3 The Schaefer Model of Fish Growth

From Population Dynamics to Growth Rate

The **Schaefer (1957) model** relates growth to the **growth function** $G(S)$ rather than the population trajectory $P(t)$. This is particularly useful for fisheries management.

The Schaefer Growth Equation

$$G(S) = rS\left(1 - \frac{S}{K}\right)$$

where: $G(S)$ = growth (flow) of fish biomass, g = intrinsic growth rate (natural growth rate when S small), S = current stock level, K = carrying capacity

Understanding the Model Components

The term $\left(1 - \frac{S}{K}\right)$ is the **key innovation**:

Stock Level	Corresponding Factor	Effect
$S \approx 0$	$(1 - \frac{0}{K}) \approx 1$	Growth near intrinsic rate
$S = K/2$	$(1 - \frac{K/2}{K}) = 0.5$	Maximum total growth rate
$S = K$	$(1 - \frac{K}{K}) = 0$	Growth stops (carrying capacity)
$S > K$	$(1 - \frac{S}{K}) < 0$	Negative growth (decline)

The Actual Growth Rate

The **actual growth rate** (growth per unit stock) is:

$$\frac{G(S)}{S} = r\left(1 - \frac{S}{K}\right)$$

This declines linearly from g when $S = 0$ to 0 when $S = K$.

Example 3.2: Analyzing Fish Stock Dynamics

Consider a fishery with $g = 0.4$ per year and $K = 10000$ tonnes.

Calculate growth at various stock levels:

Stock S (tonnes)	$\frac{S}{K}$	Calculation	Growth Rate $G(S)/S$
2,000	0.20	$0.4 \times (0.8) = 0.32$	32.0%
5,000	0.50	$0.4 \times (0.5) = 0.20$	20.0%
7,500	0.75	$0.4 \times (0.25) = 0.10$	10.0%
9,000	0.90	$0.4 \times (0.10) = 0.04$	4.0%
10,000	1.00	$0.4 \times (0.0) = 0.0$	0.0%
11,000	1.10	$0.4 \times (-0.10) = -0.04$	-4.0%

Key Observation: Growth is maximum at $S = 4000$ (where stock is exactly $K/2$).

3.4 Sustainable Yield (MSY)

The Concept

Maximum Sustainable Yield (MSY) is the largest harvest that can be taken from a renewable resource indefinitely without depleting the stock.

Figure 2: Schaefer Model and MSY

At MSY: Harvest = Growth. Stock remains constant year after year.

Finding the MSY

The Schaefer model is a parabola ($G(S) = aS - bS^2$), so the maximum occurs at $S = K/2$ (the intercept-opening parabola).

Three ways to find MSY:

1. **Use calculus:** Write $Y = G(S)$, $\frac{dY}{dS} = 0$
2. **Use the vertex formula:** For $G(S) = -\frac{g}{K}S^2 + gS$, the vertex is at $S = \frac{K}{2}$

Using the vertex formula

Rewriting $G(S)$: $G(S) = -\frac{g}{K}S^2 + gS$

Standard form: $G(S) = aS^2 + bS$ where $a = -\frac{g}{K}$ and $b = g$

Vertex at:

$$S^* = -\frac{b}{2a} = -\frac{g}{-2\frac{g}{K}} = \frac{K}{2}$$

MSY occurs when the stock is at half the carrying capacity:

$$S_{MSY} = \frac{K}{2}$$

The MSY harvest is:

$$G_{MSY} = r\frac{K}{2}\left(1 - \frac{K/2}{K}\right) = r\frac{K}{2}\left(1 - \frac{1}{2}\right)$$

$$G_{MSY} = \frac{rK}{4}$$

Example 3.3: MSY Calculation

For $g = 0.5$ and $K = 4000$ tonnes:

$$S_{MSY} = \frac{4000}{2} = 2000 \text{ tonnes}$$

$$G_{MSY} = \frac{0.5 \times 4000}{4} = 500 \text{ tonnes}$$

Interpretation: By maintaining the fish stock at 2,000 tonnes, we can sustainably harvest 500 tonnes every year—forever.

Management Implications

Current Stock	Management Action
$S < S_{MSY}$	Reduce harvest to let stock recover
$S = S_{MSY}$	Harvest at MSY rate to maintain steady state
$S > S_{MSY}$	Can harvest more than S_{MSY} temporarily to reduce stock to optimal level

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3.5 Connecting Models: Exponential → Logistic

The Differential Equation View

The Schaefer model can also be written as a differential equation:

$$\frac{dS}{dt} = gS\left(1 - \frac{S}{K}\right)$$

When $S \ll K$, this simplifies to:

$$\frac{dS}{dt} \approx gS$$

which gives exponential growth $S(t) = S_0e^{gt}$.

The solution to the full logistic differential equation is:

$$S(t) = \frac{C}{1 + Ae^{-rt}}$$

where $A = \frac{C - S_0}{S_0}$.

The Schaefer growth model and the logistic function describe the same phenomenon—one focuses on growth rate, the other on the population trajectory.

3.6 Arithmetic Sequences: The Linear Exception

Definition

While geometric sequences connect to exponential functions, **arithmetic sequences** connect to **linear functions**.

A sequence $\{a_n\}_{n=1}^{\infty}$ is **arithmetic** if consecutive terms differ by a constant:

$$a_n = a_1 + (n - 1)d$$

where: a_1 = first term, d = **common difference**, n = term number

Examples

Sequence	a_1	d	Formula
2, 5, 8, 11, 14, ...	2	3	$a_n = 2 + 3(n - 1) = 3n - 1$
100, 90, 80, 70, ...	100	-10	$a_n = 100 - 10(n - 1) = 110 - 10n$
5, 5, 5, 5, ...	5	0	$a_n = 5$

Connection to Linear Functions

An arithmetic sequence is a **discrete sampling** of a linear function:

$$a_n = a_1 + (n - 1)d = f(n) \text{ where } f(x) = a_1 + (x - 1)d$$

Compare with slope-intercept form $y = mx + b$. Slope $m = d$ (the common difference). The sequence values lie exactly on the line at integer points.

Example 3.4: Fish Stock Under Constant Decline

Suppose fish stock declines linearly due to environmental degradation:

$$S(n) = 5000 - 150n$$

where S_n = stock in year n.

Questions: 1. What is the initial stock? 2. What is the common difference? 3. When will the stock reach zero?

Solution:

1. Initial stock: $a_1 = 5000 - 150(1) = 4850$ tonnes
2. Common difference: $d = -150$ tonnes/year (declining)
3. Stock reaches zero when:

$$5000 - 150(n - 1) = 0$$
$$n - 1 = 33$$
$$n = 33 \text{ years}$$

Arithmetic Series

The sum of the first n terms of an arithmetic sequence:

$$S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n - 1)d)$$

3.7 Function Composition

Definition

Given two functions f and g, the **composition** $f \circ g$ is defined as:

$$(f \circ g)(x) = f(g(x))$$

Read as: "f of g of x." Are apply g then apply f to the result.

Important Notes

- **Order matters:** Generally, $f \circ g \neq g \circ f$
- **Domain consideration:** x must be in the domain of g and $g(x)$ must be in the domain of f

Example 3.5: Composing Functions

Let $f(x) = x^2$ and $g(x) = 3x + 5$.

(a) Find $(f \circ g)(x)$:

$$(f \circ g)(x) = f(g(x)) = f(3x + 5) = (3x + 5)^2$$

(b) Find $(g \circ f)(x)$:

$$(g \circ f)(x) = g(f(x)) = g(x^2) = 3x^2 + 5$$

(c) Evaluate $(f \circ g)(2)$:

$$(f \circ g)(2) = (3(2) + 5)^2 = 11^2 = 121$$

Example 3.6: Composition with Logistic Function

Let $f(x) = \frac{1}{1+x}$. Logistic function and $g(x) = 2x$.

$$(f \circ g)(x) = f(2x) = \frac{1}{1+2x}$$

This is exactly the dosage-risk function from Week 2 Example 2.9!

Application: Building Complex Models

In scientific modeling, composition allows us to build complex models from simpler components:

$$\text{Risk} = f(\text{dosage})$$
$$\text{Dosage} = g(\text{patient})$$
$$\text{Risk} = (f \circ g)(\text{patient})$$

3.8 Inverse Functions

Definition

If f is a one-to-one function, its **inverse** f^{-1} satisfies:

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(x)) = x$$

Graphically: The graph of f^{-1} is the reflection of the graph of f across the line $y = x$.

Finding Inverse Functions

Step 1: Write $y = f(x)$

Step 2: Solve for x in terms of y

Step 3: Swap x and y to get $y = f^{-1}(x)$

Example 3.7: Finding Inverses

(a) Find the inverse of $f(x) = 3 + \frac{1}{x}$:

Step 1: $y = 3 + \frac{1}{x}$

Step 2: Solve for x:

$$y - 3 = \frac{1}{x}$$
$$x = \frac{1}{y - 3}$$

Step 3: Swap x and y:

$$f^{-1}(x) = \frac{1}{x - 3}, \quad x \neq 3$$

Verification:

$$f(f^{-1}(x)) = f\left(\frac{1}{x-3}\right) = 3 + \frac{1}{\frac{1}{x-3}} = 3 + (x - 3) = x$$

(b) Find the inverse of $f(x) = x^3$:

Step 1: $y = x^3$

Step 2: Take cubed-log

$$\log y = 3\log x$$
$$x = \sqrt[3]{y}$$

Step 3: $f^{-1}(x) = \sqrt[3]{x} = \frac{1}{3}\log x$

Inverse of the Logistic Function

For the logistic function $P = \frac{C}{1+Ae^{-rt}}$, we can solve for t:

$$P(1 + Ae^{-rt}) = C$$
$$1 + Ae^{-rt} = \frac{C}{P}$$
$$Ae^{-rt} = \frac{C}{P} - 1$$
$$e^{-rt} = \frac{C - P}{AP}$$
$$-rt = \ln\left(\frac{C - P}{AP}\right)$$
$$t = -\frac{1}{r}\ln\left(\frac{C - P}{AP}\right)$$

This tells us **time required to reach a given population level**.

3.9 Why This Matters: Preparing for Optimization

Next week we'll combine bounded growth and the Schaefer model to extract programs for:

Prices Taker	Competitive
Week 4-5: Derivatives	Finding MSP using $\frac{dG}{dS} = 0$
Week 6: Optimization	Maximizing sustainable yield
Week 7: Exponentials	Understanding why growth slows in interacting populations
Week 14: Linear Programming	Resource allocation under constraints

The Schaefer model demonstrates how **quadratic structure** (from the product $P \times (K/K)$) leads to **optimization problems** with clear meaning—a theme that will recur throughout the course.

Summary: Key Formulas for Week 3

Topic	Key Formula
Logistic function	$P(t) = \frac{C}{1+Ae^{-rt}}$
Logistic function (alt)	$P(t) = \frac{L}{1+e^{-k(t-t_0)}}$
Schaefer growth model	$G(S) = rS\left(1 - \frac{S}{K}\right)$
MSY stock level	$S_{MSY} = \frac{K}{2}$
MSY harvest	$G_{MSY} = \frac{rK}{4}$
Arithmetic sequence	$a_n = a_1 + (n - 1)d$
Arithmetic series	$S_n = \frac{n}{2}(2a_1 + (n - 1)d)$
Function composition	$(f \circ g)(x) = f(g(x))$
Inverse relationship	$f^{-1}(f(x)) = x$

Materials adapted from ECON1340 lecture notes, Zhou, Burke and Jagnani with ECON1340 Sample Final Examinations.